Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 04** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer question (2 marks) and 3 Short Answer questions (3 marks each).
4. **Section C** comprises 1 Applied Case Study question of 4 marks with internal choice.
5. Use of calculators is not permitted.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. The 4th term from the end of the AP $-11, -8, -5, \ldots, 49$ is: [1]

(A) 37 (B) 40 (C) 43 (D) 58

Q2. The famous mathematician associated with finding the sum of the first 100 natural numbers is: [1]

(A) Pythagoras (B) Newton (C) Gauss (D) Euclid

Q3. If the first term of an AP is -5 and the common difference is 2, then the sum of the first 6 terms is: [1]

(A) 0 (B) 5 (C) 6 (D) 15

Q4. The sum of the first 16 terms of the AP $10, 6, 2, \ldots$ is: [1]

(A) -320 (B) 320 (C) -352 (D) -400

Q5. **Assertion (A):** Three consecutive terms a, b, c are in AP if and only if $2b = a + c$.
Reason (R): In an AP, the difference between consecutive terms is equal: $b - a = c - b$. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Short Answer Questions (11 Marks)

Q6. Find the sum of the first 22 terms of the AP $8, 3, -2, \ldots$. [2]

Q7. If S_1, S_2, S_3 be the sum of $n, 2n, 3n$ terms respectively of an AP, prove that: [3]

$$S_3 = 3(S_2 - S_1)$$

Q8. Find four numbers in AP whose sum is 20 and the sum of whose squares is 120. [3]

Q9. An AP consists of 37 terms. The sum of the three middlemost terms is 225 and the sum of the last three terms is 429. Find the AP. [3]

Q10. CASE STUDY: Construction Site Timber Log Stacking



At a civil infrastructure construction site, heavy cylindrical timber logs are delivered to support shoring works. To prevent accidental rolling and optimize storage space, the site supervisor orders the logs to be stacked in a triangular profile.
A total of 200 logs are stacked such that there are 20 logs in the bottom row, 19 logs in the row directly above it, 18 in the next row, and so on, reducing by 1 log in each successive layer.

Based on the above mathematical scenario, answer the following questions:

(i) Write down the values of the first term a and common difference d for the AP representing the number of logs from bottom to top. **[1]**

(ii) How many logs are placed in the 8th row from the bottom? **[1]**

(iii) In how many rows are all 200 logs placed, and how many logs are in the topmost row? **[2]**

— OR —

What is the total combined number of logs placed in the top two rows? **[2]**

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